

Quick Check

Lesson 10: Final Model and Return to the Phenomenon: How Does the Body Make the Billions of New Cells Needed for Growth?

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.4: Cell Division | Lesson 10

A baby of about 3 kg grows into a teenager of 50 kg. A student says, "Cells just get bigger as we eat ugali, githeri and sukuma wiki." Which statement best explains most of the growth?

- A** The number of cells increases because existing cells divide to make new cells
- B** Food turns directly into new cells inside the body
- C** Cells from the food we eat are absorbed and added to the body
- D** Cells only get bigger, and their number stays the same

A human muscle cell has 46 chromosomes. It divides by mitosis. What is produced?

- A** Two cells, each with 23 chromosomes
- B** Four cells, each with 23 chromosomes
- C** Two cells, each with 46 chromosomes that are identical to the parent cell's
- D** Two cells, each with 46 chromosomes but with different instructions from each other

A zygote (1 cell) divides by mitosis. Every cell then divides once in each round, so the pattern is 1, 2, 4, 8, 16... How many cells are there after 5 rounds of division?

A 25

B 10

C 32

D 16

Yeast cells in mandazi dough and bacteria in milk both reproduce by dividing. Which statement correctly distinguishes budding from binary fission?

- A** Budding splits a cell into two equal halves; binary fission forms a small outgrowth on the parent cell
- B** Budding happens only in bacteria and binary fission happens only in yeast
- C** Both budding and binary fission make four cells with half the chromosomes
- D** Budding is an unequal division in which a small bud grows from a yeast cell and breaks off; binary fission splits a bacterium into two equal cells

Before a body cell divides by mitosis, its DNA is copied during interphase. Why is this important?

- A** So that food can be changed into new cells
- B** So that each daughter cell receives only half of the instructions
- C** So that the cell can become bigger before dividing
- D** So that each of the two daughter cells receives a full set of identical instructions

Banana plantlets are grown in a laboratory from one small piece of banana tissue. Which idea from cell division does this show?

- A** Plants can only form from the joining of two gametes
- B** The piece of tissue grows bigger but its cells never divide
- C** Cells from one piece of tissue can divide repeatedly by mitosis and form a whole new plant
- D** Meiosis in the tissue makes many new plantlets

A group's final model of how a child grows says, "Meiosis makes the new muscle and bone cells needed for growth." What correction should the group make?

- A** Meiosis is correct, because it makes four identical body cells
- B** Growth happens only by cells getting bigger, so meiosis and mitosis should be removed
- C** Growth happens by binary fission, which makes the new muscle and bone cells
- D** Growth happens by mitosis; meiosis makes gametes with half the chromosome number

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).